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ST10507000

PDAN8411 Assignment 1 Part 1

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Contents

Introduction 2
1. Evaluating the dataset 2
2. The analysis plan and results 3
a. Exploratory Data Analysis (EDA) 3
b. Feature selection 6
c. Train model 8
d. Interpret and evaluate model 9
5. Improving the model 10
References 11

ST10507000_Part 1 – Linear Regression Report

Introduction

This assignment investigates how linear regression can predict medical insurance costs. The (Choi, 2017) dataset contains numerical variables which include age and BMI and children and charges and it contains categorical variables which include sex and smoker and region. The analysis starts with data exploration which includes summary statistics and correlation analysis to find possible predictor variables. Categorical variables are encoded using a combination of binary encoding (for sex and smoker) and one-hot encoding (for region). The process of feature selection depends on two factors which are correlation strength and statistical significance (p-values) and it uses backward elimination to keep only essential features. The model is tested through R^2 and Mean Absolute Error (MAE) and Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) evaluation metrics. The researchers test various enhancements to linear regression which include interaction terms and polynomial terms and log transformation to improve prediction accuracy while preserving model interpretability.

1. Evaluating the dataset

for suitability to apply a Linear Regression algorithm.

The Plan	
The dataset was checked for suitability using the below measures for Linear Regression:	
Suitability measure	**Measure detail**
Continuous target variable	The target variable (charges) is continuous, which is required for regression models.
Multiple independent variables	The dataset contains multiple independent variables (age, sex, BMI, children, smoker, region) that can be used to predict the dependent variable.
Missing data	There is no indication of excessive missing data, making it appropriate for modelling after basic cleaning.
Linear relationships	Relationships between variables such as age, BMI, and smoking status appear to have a linear/ near-linear trend with charges/cost.
Dataset size	The dataset contains 1300 rows making it sufficient for a basic regression model.
No time dependency	Data is cross-sectional seems to be no need for time-series models.
Categorical variables	Categorical variables (sex, smoker, region) can be encoded into numerical form.
The results	
	The examined the dataset to determine its compatibility with linear regression analysis. The target variable medical charges exists as a continuous measurement which multiple independent variables provide through their complete input, which includes age, BMI, number of children, sex, smoking status, and region information which provide sufficient information for prediction (Montgomery et al., 2021). The dataset enables the numerical representation of categorical variables (Géron, 2019) because it contains more than 1300 data entries, which are sufficient to create a basic regression model. The exploratory analysis shows that multiple predictors demonstrate almost direct relationships, but smoking displays non-linear effects, which show how it affects charges. Standard regression techniques become suitable because the data does not exhibit any time-dependent patterns. The dataset remains clean because it contains only minor missing data, but its analysis shows three main issues which include non-linear pattern detection and variable

	interaction and restricted results to the original population studied in the United States. The dataset becomes appropriate for linear regression analysis after researchers complete preprocessing and encoding processes while they assess the established limitations.

2. The analysis plan and results

a. Exploratory Data Analysis (EDA)

The Plan
To understand structure, detect issues, and identify patterns.

Step	Detail and results	
1	**Step detail**	Load dataset in Jupyter using Pandas and inspect structure using Pandas functions like .head() and .info()

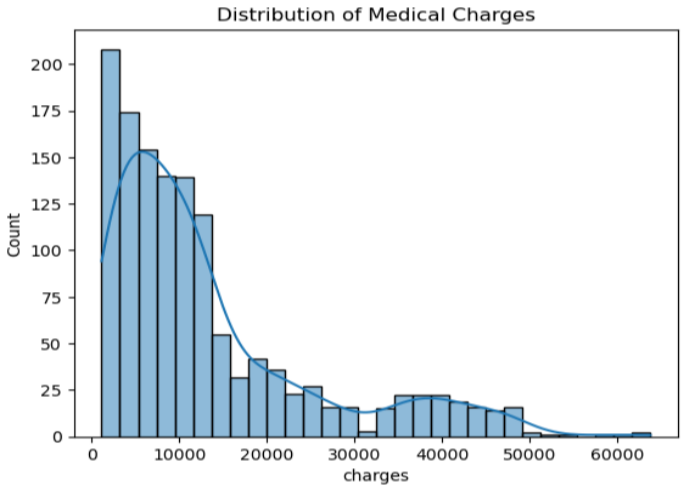
	Results	View few first rows
		`print(df.head())`
		`  age  sex  bmi  children smoker  region  charges`
		`0  19  female  27.900  0  yes  southwest  16884.92400`
		`1  18  male  33.770  1  no  southeast  1725.55230`
		`2  28  male  33.000  3  no  southeast  4449.46200`
		`3  33  male  22.705  0  no  northwest  21984.47061`
		`4  32  male  28.880  0  no  northwest  3866.85520`
		Check structure
		`print(df.info())`
		`<class 'pandas.core.frame.DataFrame'>`
		`RangeIndex: 1338 entries, 0 to 1337`
		`Data columns (total 7 columns):`
		`#  Column  Non-Null Count  Dtype`
		`0  age  1338 non-null  int64`
		`1  sex  1338 non-null  object`
		`2  bmi  1338 non-null  float64`
		`3  children  1338 non-null  int64`
		`4  smoker  1338 non-null  object`
		`5  region  1338 non-null  object`
		`6  charges  1338 non-null  float64`
		`dtypes: float64(2), int64(2), object(3)`
		`memory usage: 73.3+ KB`
		`None`
		The dataset was successfully loaded and explored using Pandas, NumPy, Matplotlib, and Seaborn for data analysis and visualization. The dataset contains 1338 entries and 7 variables which constitutes a dataset that researchers can analyse because of its moderate size. The dataset contains numerical variables which include age and BMI and children and charges together with categorical variables which include sex and smoker and region.

2	**Step detail**	Check for any missing values and also check for duplicates. Any duplicates found must be removed to avoid bias in the model.

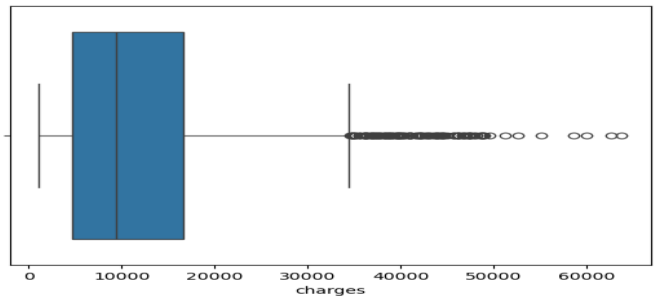
	Results	Check missing values
		`print(df.isnull().sum())`
		`age  0`
		`sex  0`
		`bmi  0`
		`children  0`
		`smoker  0`
		`region  0`
		`charges  0`
		`dtype: int64`
		Check duplicates
		`print(df.duplicated())`
		`0  False`
		`1  False`
		`2  False`
		`3  False`
		`4  False`
		`...`
		`1333  False`
		`1334  False`
		`1335  False`
		`1336  False`
		`1337  False`
		`Length: 1338, dtype: bool`
		There are no missing values, as all columns have 1338 non-null entries, and no visibility of any duplicates meaning no immediate data cleaning is required for missing or duplicate data entries (McKinney, 2022).

4	**Step detail**	Summary statistics

	Results	Check summary statistics `print(df.describe())` The summary statistics show that the average age is about 39 years and the average BMI is around 30.7. Most individuals have about 1 child. Medical charges vary widely, indicating high variability and possible outliers (James *et al*., 2021).
5	**Step detail**	Visualise distributions using the histograms

	Results	Distribution of charges
		`sns.histplot(df['charges'], kde=True)`
		`plt.title("Distribution of Medical Charges")`
		`plt.show()`
		 
		
		The data is said to be from the US as such the amounts will be depicted as dollars. The distribution of medical charges is right-skewed, with most individuals incurring relatively low charges under $15,000. There is a long tail extending toward the higher charges on the graph, which indicating some individuals have very high medical costs, thus making it to aligns directly with the wide range and potential outliers seen in the summary statistics shown above.

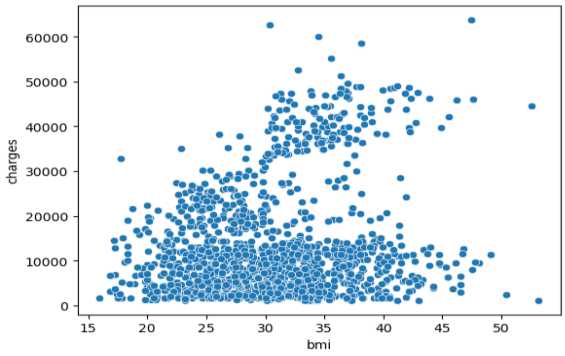
6	**Step detail**	Detect any outliers with boxplots more especially in BMI and charges

	Results	Boxplot for outliers
		`sns.boxplot(x=df['charges'])`
		`plt.show()`
		 
		
		The boxplot shows that most medical charges are concentrated below around $35,000, but there are many outliers extending beyond this point in the graph. These outliers identified using charges in the boxplot represent individuals with unusually high medical costs compared to the majority of the dataset.

7	**Step detail**	Analyse relationships using scatter plots and correlation matrix (heatmap).
	Results	

Relationship: BMI with Charges

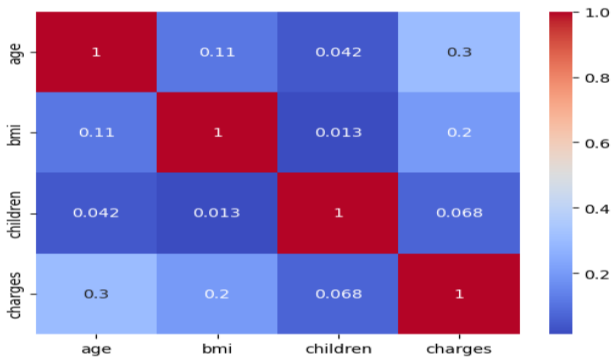
```
sns.scatterplot(x='bmi', y='charges', data=df)
plt.show()
```



The relationship shown in the scatter plot indicates that with higher BMI that would indicate obesity comes higher medical charges but even with this visibility of trend the plots are somewhat widely spread in some instances which would normally indicate that that BMI does not fully influence the differences of medical charges alone.

Correlation heatmap

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.show()
```

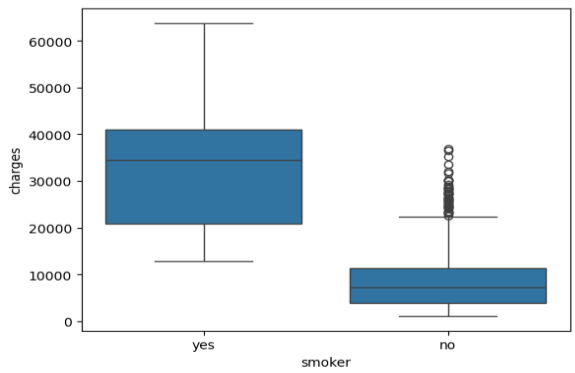


The heatmap is showing us the strength and direction of the relationship between each numerical variables (age, BMI, children, charges) with the strongest one being age and charges with a 0.30 mark. With the weakest correlation of BMI with charges of 0.20 mark proving a increase of charges as BMI increases. The correlation also indicates a moderate positive of the older the individuals the likelihood to have higher medical charges. Then on the numerical variable children there seems to be no relationship at all with a very weak 0.068.

8	**Step detail**	Compare categorical variables using box plots the smoker against charges.
	Results	

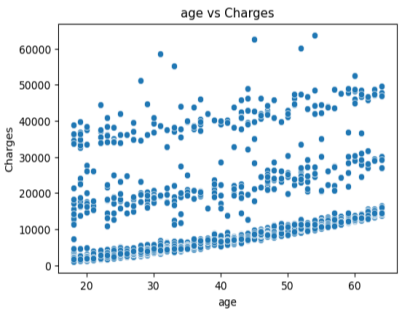
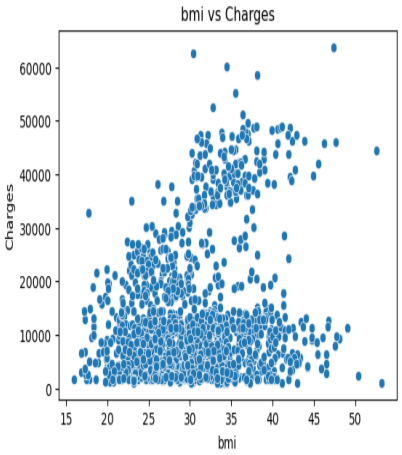
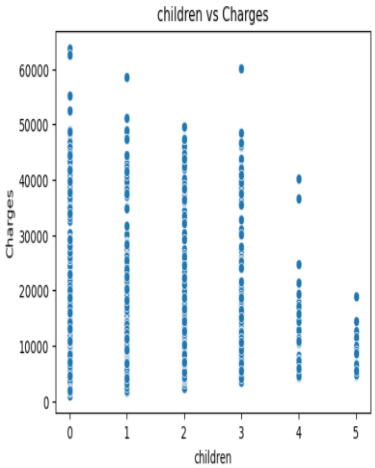
Smoking impact: Compare categorical variables using box plots (smoker vs charges)

```
sns.boxplot(x='smoker', y='charges', data=df)
plt.show()
```



In step 1 of The EDA analysis we identified the categorical variables (sex, smoker, region). The comparison in this box plot was done using smoker against charges which show that smokers have high charges than those that do not smoke. And the plot also shows a visibility in some outliners. The plot displays that smoking as a categorical variable has a strong relationship with charges

9	**Step detail**	Relationship with target (scatter plots)

	Results	Relationship with target (scatter plots)
		`# Scatter plots for numerical variables vs charges`
		`numerical_cols = ['age', 'bmi', 'children']`
		`for col in numerical_cols:`
		`  plt.figure(figsize=(6,4))`
		`  sns.scatterplot(x=col, y='charges', data=df)`
		`  plt.title(f'{col} vs Charges')`
		`  plt.xlabel(col)`
		`  plt.ylabel('Charges')`
		`  plt.show()`
		 
		 
		 
		Indication of a clear positive relationship showing that as the older the age the higher the charges
		The relationship is weak as the graph indicates a non-linear pattern of lower charges but a higher BMI showing an association with high charges.
		There seems to be no relationship at all as charges are just spread across all of the children showing that the number of children has no effect on medical charges.
		With the three relationships comparison you can see that age is the strongest predictor amongst the three and BMI does have some influence with some outliners and children has no impact at all to the medical charges.

b. Feature selection

The plan

To be able to get the correct features for the linear regression model, all the variables initially will all be considered (Kutner *et al*., 2018).
• Will be encoding categorical variables a combination of one-bot Encoding alongside binary encoding
  – Sex will be binary (0/1)
  – Smoker will be binary
  – Region will be one-hot encoding
• Use correlation matrix to identify the strongest predictors amongst the variables
• Will apply statistical methods such as p-values to see feature significance amongst the variables
• Will be using backward or forward elimination to remove any irrelevant variables amongst the variables
 	
The deliverable for the plan is to be able to keep only from the dataset the strongest predictor amongst the variables

The results

Step : Encode categorical variables

```python
# Sex will be binary (0/1)
df['sex'] = df['sex'].map({'male': 0, 'female': 1})
# Smoker will be binary (0/1)
df['smoker'] = df['smoker'].map({'no': 0, 'yes': 1})
# Region will be one-hot encoding
df = pd.get_dummies(df, columns=['region'], drop_first=True, dtype=int)
# View first few rows
print(df.head())
```

```
   age  sex     bmi  children  smoker      charges  region_northwest  \
0   19    1  27.900         0       1  16884.92400                 0
1   18    0  33.770         1       0   1725.55230                 0
2   28    0  33.000         3       0   4449.46200                 0
3   33    0  22.705         0       0  21984.47061                 1
4   32    0  28.880         0       0   3866.85520                 1

   region_southeast  region_southwest
0                 0                 1
1                 1                 0
2                 1                 0
3                 0                 0
4                 0                 0
```

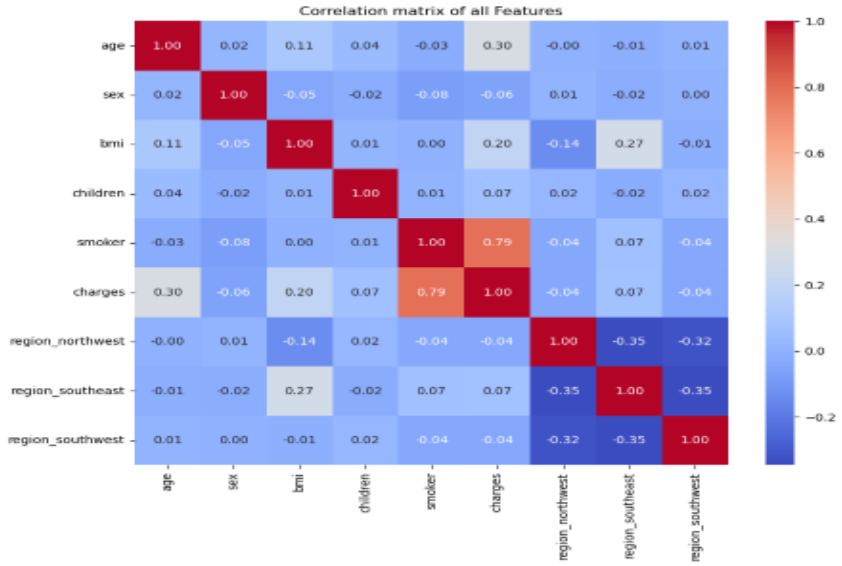
Step: correlation matrix to identify the strongest predictors amongst our variables

```python
corr_matrix = df.corr()

# correlation with target variable 'charges'
print("Correlation with medical charges:")
print(corr_matrix['charges'].sort_values(ascending=False))

# the correlation matrix map
plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, fmt=".2F", cmap='coolwarm')
plt.title('Correlation matrix of all Features')
plt.show()
```

```
Correlation with medical charges:
charges             1.000000
smoker              0.787251
age                 0.299008
bmi                 0.198341
region_southeast    0.073982
children            0.067998
region_northwest   -0.039905
region_southwest   -0.043210
sex                -0.057292
Name: charges, dtype: float64
```



Each categorical variable was converted to sex as a binary as well as smoker as a binary making it either 0/1 and the region has been applied as a one hot encoded column, creating new binary columns such as region_northwest, region_southeast, and region_southwest making the dataset to be ready for any regression and feature selection

To define the predictors against the target a correlation matrix was used to identify the strongest predictor amongst the variables. The heat map interpretation is that the closer the variable is to +1 the stronger the positive correlation, the closer to -1 the strong negative the variable is in correlation while close to 0 indicates no relation at all. For manual feature selection we can see that the strong positive correlation is smoker, age which qualifies them as the strongest predictors. BMI appearing as moderate but would still matter as a strong predictor. Children, sex, and the region dummies are showing indicate near zero correlation which results as no importance at all.

After correlation we move to statistical significance using OLS regression and backward elimination (Montgomery *et al*., 2021).

Statistical significance using p-values

Backward elimination.

Step : statistical methods such as p-values to see feature significance amongst the variables

```python
# Defining predictors (X) and target (y)
X = df.drop('charges', axis=1)
y = df['charges']

# Adding a constant term for the intercept
X = sm.add_constant(X)

# Fitting full OLS model to check p-values
full_model = sm.OLS(y, X).fit()
print("Full model summary:\n")
print(full_model.summary())
```

```
Full model summary:

                            OLS Regression Results
==============================================================================
Dep. Variable:                charges   R-squared:                       0.751
Model:                            OLS   Adj. R-squared:                  0.749
Method:                 Least Squares   F-statistic:                     500.8
Date:                Tue, 31 Mar 2026   Prob (F-statistic):               0.00
Time:                        13:22:55   Log-Likelihood:                -13548.
No. Observations:                1338   AIC:                         2.711e+04
Df Residuals:                    1329   BIC:                         2.716e+04
Df Model:                           8
Covariance Type:            nonrobust
====================================================================================
                       coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------
const            -1.207e+04    999.649    -12.074      0.000    -1.4e+04   -1.01e+04
age                256.8564     11.899     21.587      0.000     233.514     280.199
sex                131.3144    332.945      0.394      0.693    -521.842     784.470
bmi                339.1935     28.599     11.860      0.000     283.088     395.298
children           475.5005    137.804      3.451      0.001     205.163     745.838
smoker            2.385e+04    413.153     57.723      0.000     2.3e+04    2.47e+04
region_northwest  -352.9639    476.276     -0.741      0.459   -1287.298     581.370
region_southeast -1035.0220    478.692     -2.162      0.031   -1974.097     -95.947
region_southwest  -960.0510    477.933     -2.009      0.045   -1897.636     -22.466
==============================================================================
Omnibus:                      300.366   Durbin-Watson:                   2.088
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              718.887
Skew:                           1.211   Prob(JB):                    7.86e-157
Kurtosis:                       5.651   Cond. No.                         315.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Features with high p-values (> 0.05) will be removed using the backward before training the model as they will add no value into explaining the medical charges the lower the p-value the significant the variable. From the results sex shows p value of 0.693 indicating no significance at all.

Step : Backward elimination to remove any irrelevant variables amongst the variables

```python
def backward_elimination(data, target, sl=0.05):
    features = data.columns.tolist()
    while True:
        X_with_const = sm.add_constant(data[features])
        model = sm.OLS(target, X_with_const).fit()
        p_values = model.pvalues.iloc[1:]  # exclude intercept
        max_p = p_values.max()
        if max_p > sl:
            removed_feature = p_values.idxmax()
            features.remove(removed_feature)
            print(f"Removed {removed_feature} (p={max_p:.4f})")
        else:
            break
    return features, model

selected_features, fin_model = backward_elimination(X.drop('const', axis=1), y)

print("\nThe backward elimination final values:")
print(selected_features)

print("\nModel summary:")
print(fin_model.summary())
```

```
Removed sex (p=0.6933)
Removed region_northwest (p=0.4596)
Removed region_southwest (p=0.0587)
Removed region_southeast (p=0.1363)

The backward elimination final values:
['age', 'bmi', 'children', 'smoker']

Model summary:
                            OLS Regression Results
==============================================================================
Dep. Variable:                charges   R-squared:                       0.750
Model:                            OLS   Adj. R-squared:                  0.749
Method:                 Least Squares   F-statistic:                     998.1
Date:                Tue, 31 Mar 2026   Prob (F-statistic):               0.00
Time:                        13:31:17   Log-Likelihood:                -13551.
No. Observations:                1338   AIC:                         2.711e+04
Df Residuals:                    1333   BIC:                         2.714e+04
Df Model:                           4
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -1.21e+04    941.984    -12.848      0.000    -1.4e+04   -1.03e+04
age          257.8495     11.896     21.675      0.000     234.512     281.187
bmi          321.8514     27.378     11.756      0.000     268.143     375.559
children     473.5023    137.792      3.436      0.001     203.190     743.814
smoker      2.381e+04    411.220     57.904      0.000     2.3e+04    2.46e+04
==============================================================================
Omnibus:                      301.480   Durbin-Watson:                   2.087
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              722.157
Skew:                           1.215   Prob(JB):                    1.53e-157
Kurtosis:                       5.654   Cond. No.                         292.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

The backward elimination
Removed sex which had a p value greater than 0.05 alongside the region dummies after all the eliminations we remain with age, BMI, children, smoker that provided a lower p-value which qualified them as significant to explaining medical charges. Making the dataset ready for testing.

c. Train model

The plan
The model is Multiple Linear Regression • Will split the dataset into training and testing sets using the 80/20 split logic • Wil use Linear Regression from Scikit-learn (LinearRegression() from sklearn) And apply default hyperparameters initially
The results

c. Import Machine learning libraries

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

c. split the dataset into training and testing sets using the 80/20 split logic

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

c. Create model use Linear Regression from Scikit-learn (LinearRegression() from sklearn)

```
model = LinearRegression()
```

c. Train model

```
model.fit(X_train, y_train)
```

LinearRegression

▼ Parameters

fit_intercept	True
copy_X	True
tol	1e-06
n_jobs	None
positive	False

c. Predictions

```
y_pred = model.predict(X_test)
```

Split dataset into training and testing sets using the 80/20 this helps to give a good dataset balance between model learning and the model evaluation. The 80 will give the model multiple examples to learn various patterns of seeing how the strongest predictors affect charges as the more the training data will generally produce a better model performance. The 20 used for test data is enough for a reliable data estimate of seeing the model performance. The 80/20 ratio was to also ensure that the model is trained on a separate section of the data and while also tested on unseen data. The usage of LinearRegression() model was with default hyperparameters, to make it a simple, effective model. The model was then trained using the data (X_train, y_train) to learn the relationship between the independent variables and medical charges. After training, predictions (y_pred) were generated using the test dataset (X_test). All this ensured the success training of the model indicating the capability of affecting medical charges based on the selected features

d. Interpret and evaluate model

The plan

- The evaluation metrics to be used for evaluation are:
  - R^2 (coefficient of determination) helps to check goodness of fit
  - Mean Absolute Error (MAE)
  - Mean Squared Error (MSE)
  - Root Mean Squared Error (RMSE)
- Analyse residual plots to check assumptions
- Interpret coefficients to understand feature impact

The results

Import more Machine learning libraries

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

a. The metrics used for evaluation are

- R^2 (coefficient of determination) helps to check goodness of fit
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

```
MAE: 4181.194473753647
MSE: 33596915.85136145
RMSE: 5796.284659276273
R2 Score: 0.7835929767120724
```

The model trained shows 78% of the data difference for medical charges is explained by the data which indicates the model is a good fit (James *et al.*, 2021). The results further show that MAE on average, the model's predictions differ from actual values by about 4181 units. The root mean squared error is about 5796 units which indicates a high error value which means that our dataset has potential outliners. The select features according to their p-values were the strongest predictors, if the dataset still has outliners while using the strongest predictors it indicates medical costs are by nature high which limits perfect model prediction but still opening room for improvements

5. Improving the model

The linear regression model was improved by applying a log transformation to the target variable to reduce skewness and the influence of outliers. Additionally, an interaction term between BMI and smoking status was introduced to capture the combined effect of these variables. These improve the ability of the model to locate the underlying patterns within the linear regression setup.

Improvement and the results

Improvement 1 Log transform target (handles skewness)	Improvement 2 Add polynomial term (curve)	Improvement 3 Add interaction term (combined effect)
`# Improvement 1 Log transform target (handles skewness)` `y = np.log(df['charges'])`	`# Improvement 2 Add polynomial term (curve)` `df['bmi_sqrd'] = df['bmi'] ** 2`	`# Improvement 3 Add interaction term (combined effect)` `df['bmi_smkr'] = df['bmi'] * df['smoker']`

5. Model Improvement

reevaluate the model¶

```
mae = mean_absolute_error(y_test_actual, y_pred)
mse = mean_squared_error(y_test_actual, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test_actual, y_pred)

print("Improved Model Performance:")
print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

```
Improved Model Performance:
MAE: 4181.194473753646
MSE: 33596915.85136146
RMSE: 5796.2846592762735
R2 Score: 0.7835929767120723
```

With these improvements model inputs will be changed so the old, trained data will no longer be valid the data will need to be trained again. There are new features added the bmi_sqrd, the bmi_smkd through the polynomial and interaction term this makes the relationship between x and y different.

After introducing a log transformation, polynomial terms, and interaction effects, the model performance remained largely unchanged. The original linear model reached its maximum performance because it already showed all essential data connections which additional features failed to provide. The simpler model becomes the better choice because it enables easier understanding while maintaining equivalent prediction accuracy.

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